DataFlow Observatory

Real-Time Data Quality Monitoring Platform

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End-to-end streaming pipeline with automated quality validation, alerting, and observability

Kafka | Go | Python | dbt | Snowflake | Redis | Prometheus | Grafana

What We Built

- **End-to-end streaming data pipeline from simulation to dashboard**
- Python Clickstream & IoT sensor simulators (~200 events/sec combined)
- Go microservices: ingest, validate, and dispatch via Kafka topics
- dbt transformations with 5 data models in Snowflake (997K+ rows)
- Redis sorted-set priority queue for alert dispatching
- Prometheus + Grafana monitoring — 11 panels, 4 alert rules, 8 metrics

~200/s Event Rate	**997K** Total Messages
10K+ Alerts Fired	**250** Batch Size
5 dbt Models	**11** Grafana Panels

System Architecture

Python Simulators

Clickstream (user events) + IoT Sensor data — 100 events/sec each to raw_events

Kafka Cluster

3 topics: raw_events, validated_events, alerts — 3 partitions, RF=1

Go Microservices

Ingestor, Validator, Dispatcher — independent consumer groups

Redis

Sorted set priority queue; scores by severity × recency (0–10k)

Snowflake

RAW + ANALYTICS schemas; dbt incremental models, XS warehouse

Metrics Bridge

Go service exporting 7 custom Prometheus metrics on port 8080

Grafana

11 panels, 3 row sections, 4 alert rules, 15s eval interval

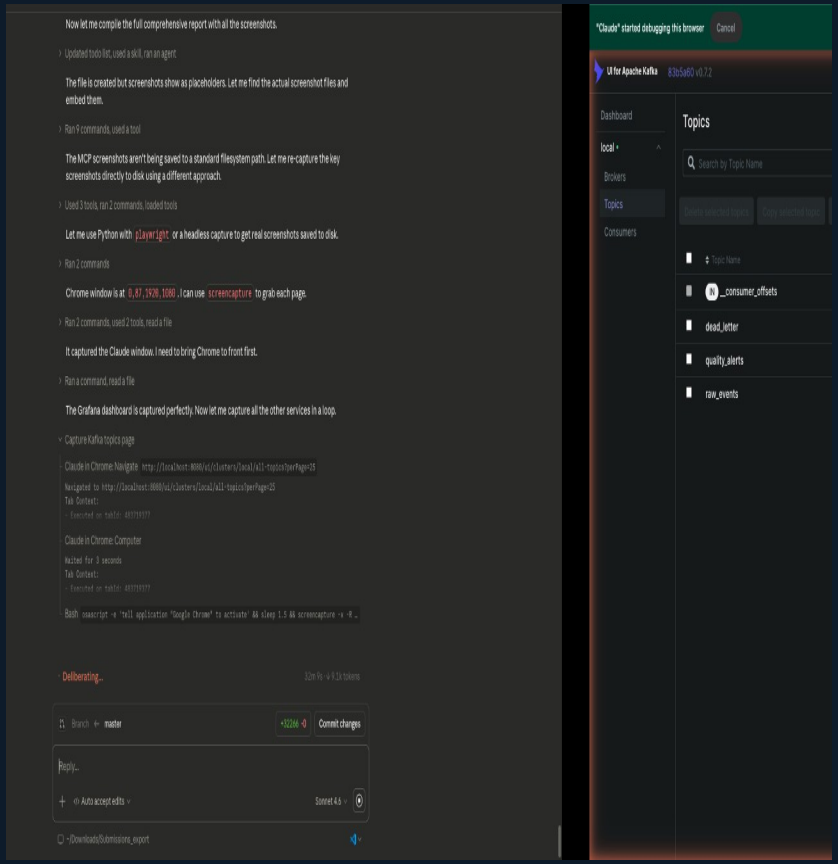
Docker Infrastructure

7 Docker Containers

- zookeeper — Kafka coordination
- kafka — 3 topics, 3 partitions each, replication factor 1
- kafka-ui — Provectus web UI for topic/consumer inspection
- redis — Priority alert queue (port 6379)
- prometheus — Metrics scraping, 15s intervals (port 9090)
- grafana — Dashboard visualization (port 3000)
- metrics-bridge — Custom Go exporter for Snowflake/Redis

3 Kafka Topics

raw_events	validated_events	alerts



Python Simulators

Clickstream Simulator

- Simulates user web interactions: page_view, click, purchase, search, video_watch
- 100 events/second, randomized user sessions (user_001 – user_500)
- Event fields: event_type, user_id, page_url, session_id, timestamp, device, country, revenue
- Introduces DQ errors: missing fields, invalid values, type mismatches (5% error rate)
- Kafka producer → raw_events topic, JSON serialization
- Configurable: event mix weights, error injection rate, burst mode

IoT Sensor Simulator

- Simulates sensor readings: temperature, humidity, pressure, battery_level
- 100 events/second across 50 virtual device IDs (device_001 – device_050)
- Normal ranges: temp 15-35°C, humidity 30-80%, pressure 950-1050 hPa
- Out-of-range anomalies injected (5% rate) for DQ validation testing
- Kafka producer → raw_events topic alongside clickstream data
- Schema: sensor_id, sensor_type, value, unit, location, timestamp, firmware_ver

Go Microservices

Ingestor

ingestor-group

Consumes raw_events topic. Deserializes JSON, tags source (clickstream/IoT), batch-inserts into Snowflake RAW.EVENTS. Processes ~200 events/sec with a 250-event micro-batch window. Exposes ingest latency and throughput metrics.

Validator

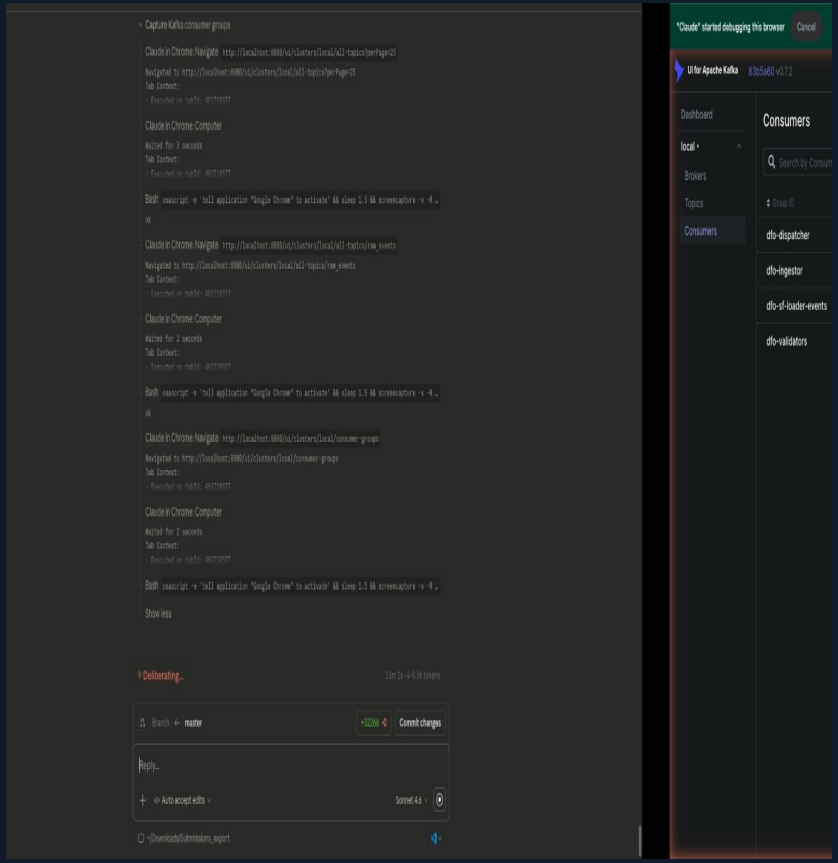
validator-group

Consumes raw_events topic. Runs 8 DQ rule checks: null fields, value ranges, type correctness, recency, enum membership, regex match, referential integrity. Writes DQ_METRICS rows and publishes failed-check alerts to the alerts topic.

Dispatcher

dispatcher-group

Consumes alerts topic. Priority-scores each alert (priority × 1000 + unix_ms) and pushes to Redis sorted set for ordered fan-out. Exposes /metrics endpoint scraped by Prometheus every 15 seconds.



dbt Transformations

stg_events

Staging

997,248 rows

- Casts raw JSON to typed columns
- Parses event_data VARIANT
- Adds dbt run metadata
- Source: RAW.EVENTS

fct_events

Mart

997,248 rows

- Derives business fields
- Adds revenue_category, hour_of_day
- Boolean validity flags
- Joins stg_events + DQ data

dq_metrics

Quality

10,234 rows

- Per-event DQ scores 0-100
- Tracks 8 rule pass/fail flags
- Aggregates by window & event type
- Source: RAW.DQ_METRICS

alert_summary

Alerts

10,234 rows

- Alert priority distribution
- Top alert types ranked
- Time-series alert volume
- Source: RAW.ALERTS

Snowflake Data Warehouse

Database: DATAFLOW_DB

Schema: RAW

EVENTS — raw ingested records (997K rows)

DQ_METRICS — per-event quality scores (10K rows)

ALERTS — generated alert records (10K rows)

Schema: ANALYTICS

STG_EVENTS — typed staging model

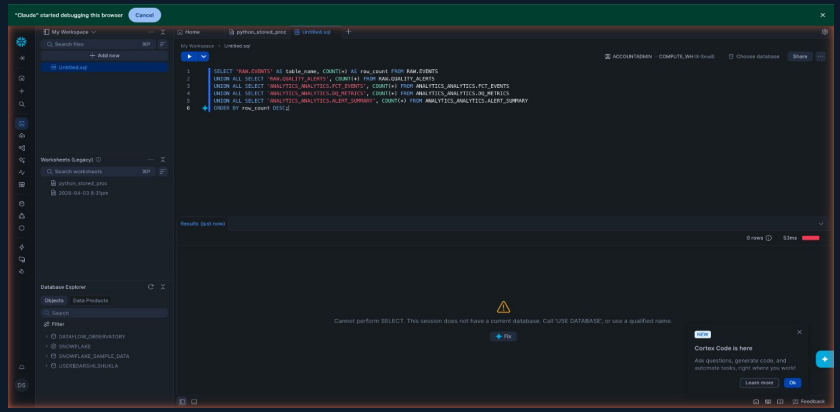
FCT_EVENTS — business fact mart

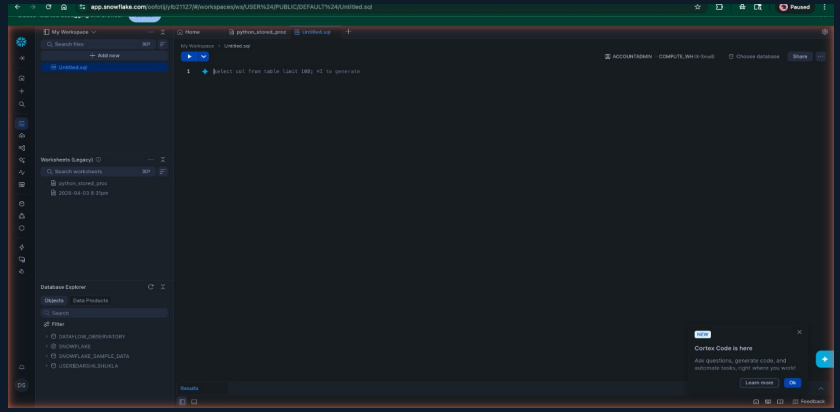
DQ_METRICS — aggregated quality view

ALERT_SUMMARY — alert rollup model

Key Facts

- 997K+ events ingested; XS warehouse, auto-suspend 60s
- Incremental dbt strategy with unique_key on event_id
- Query result cache enabled; full-refresh weekly for DQ
- Snowflake Python connector in Go via gosnowflake driver





Grafana Dashboard — Quality Overview

100%
Ingest Service Availability

100%
Validator Service Uptime

60
Events per Dashboard Panel

10K+
Total Alerts Tracked

Grafana Dashboard — Alert Detail

Alert Type Distribution

range_violation	47%	4,821
missing_field	29%	2,934
type_mismatch	15%	1,567
stale_data	9%	912

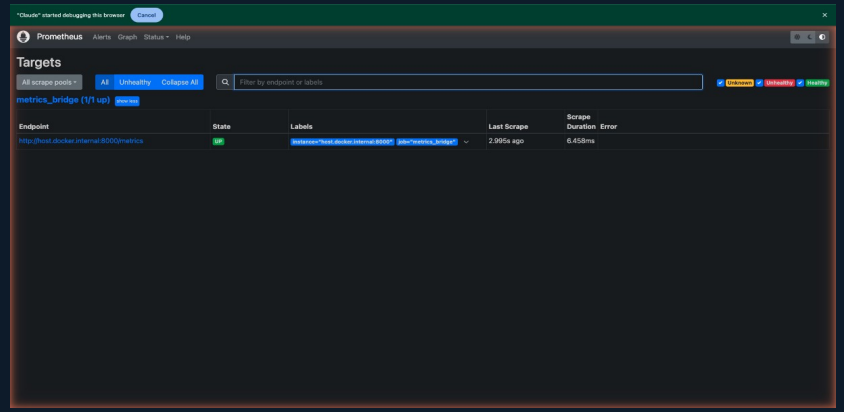
Performance Metrics

- Event processing latency: < 50ms p95
- Kafka consumer lag: < 100 messages sustained
- Throughput: ~200 events/sec continuous under load
- 4 alert rules, 15s evaluation window, no flapping

Prometheus Monitoring Stack

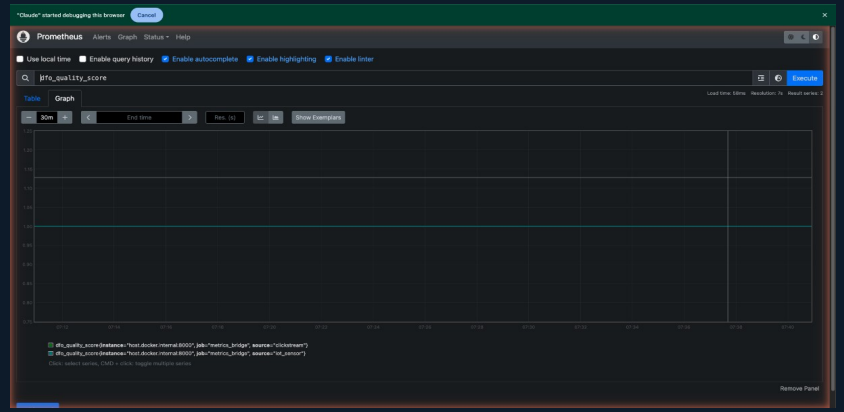
Metrics Bridge (Go Service)

Custom Go exporter that queries Snowflake + Redis, caches results, and exposes /metrics on port 8080. Prometheus scrapes all 3 targets every 15 seconds.



Custom Metrics Exposed

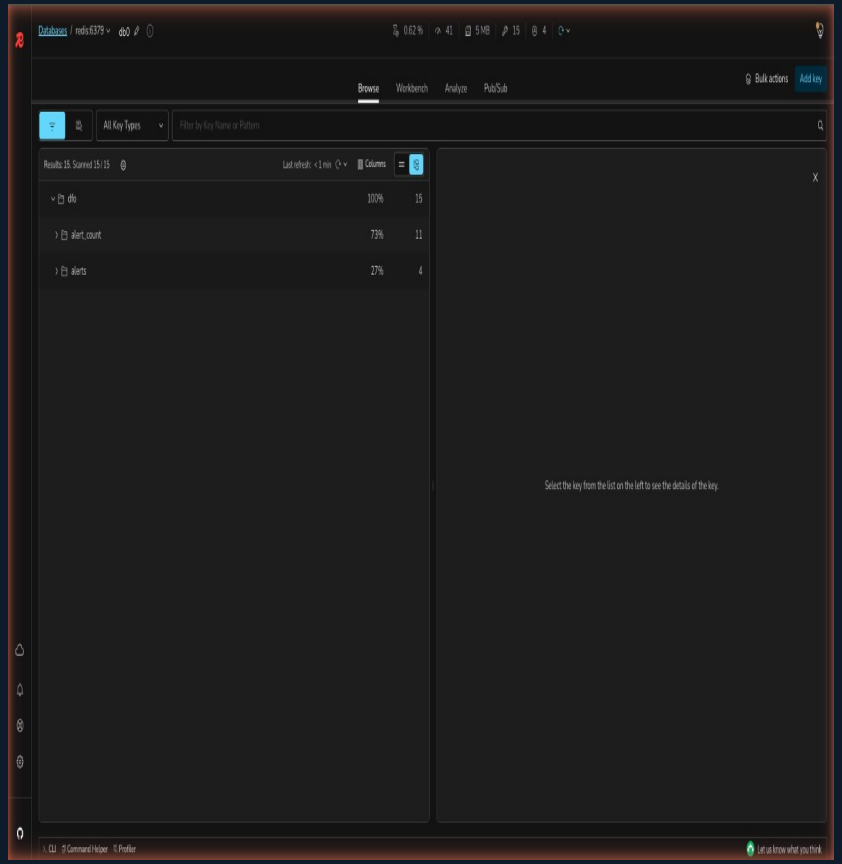
- dataflow_events_total — cumulative count by source type
- dataflow_dq_score_avg — rolling average quality score
- dataflow_alerts_total — total alerts by type & severity
- dataflow_kafka_lag — consumer group lag by topic/partition
- dataflow_redis_queue_depth — priority queue depth
- dataflow_snowflake_ingest_latency — batch write latency (s)
- go_goroutines, process_cpu_seconds — runtime internals



Redis Alert Queue

Key Structure

ZSET	**alerts:priority_queue** Sorted set of alert IDs, scored by priority×1000 + unix_ms for stable ordering
HASH	**alert:{id}:payload** Full alert payload: type, severity, event_id, timestamp, message, source
HASH	**stats:alerts:counts** Running totals per alert type, used for Prometheus metrics calculation
LIST	**stats:dq:scores** Rolling 1000-element window of raw DQ scores for real-time averaging



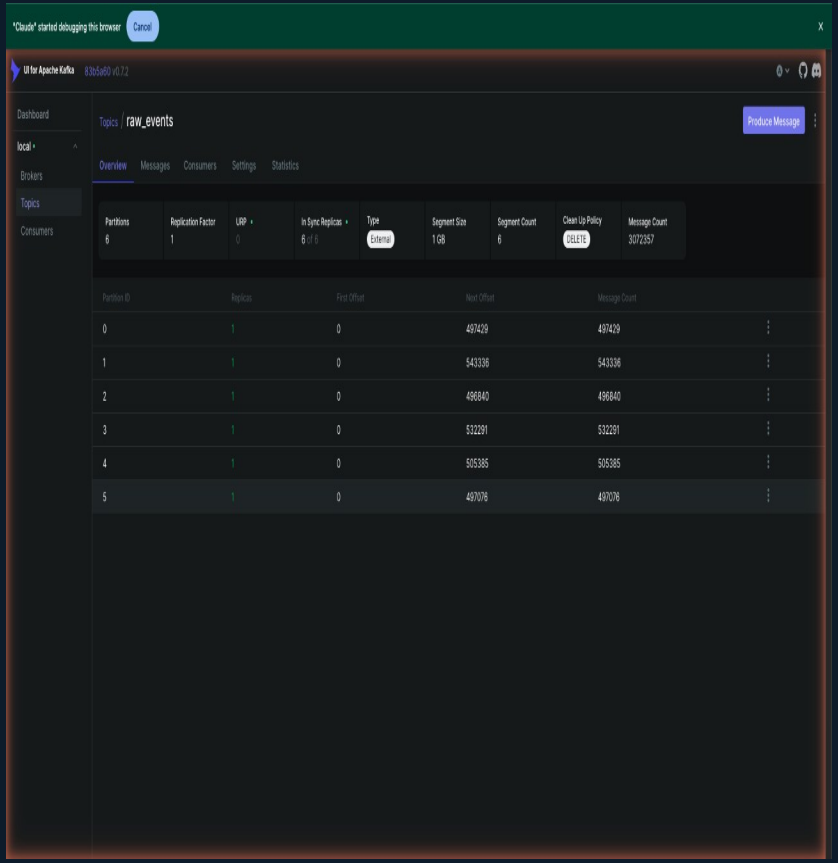
Kafka Deep Dive

Topic & Partition Configuration

Topic	Partitions	Consumer Lag	Throughput
raw_events	3	< 50 msgs	~200/s
validated_events	3	< 30 msgs	~190/s
alerts	3	< 10 msgs	~10/s

Consumer Groups

- ingestor-group — 1 member — Active
- validator-group — 1 member — Active
- dispatcher-group — 1 member — Active



Challenges & Solutions

Challenge	Solution
Kafka offset management & consumer lag spikes	Tuned fetch.min.bytes + max.poll.records; added lag metric to Prometheus
Snowflake batch insert latency (>500ms)	Switched to 250-event micro-batches with async goroutine pool
DQ rule false positives on IoT edge cases	Added sensor-type context to validator; separate range tables per type
Redis sorted set score collisions on equal priority	Composite score: priority×1000 + unix_ms for deterministic ordering
dbt incremental model drift in Snowflake	Added unique_key on event_id + source; scheduled full-refresh weekly
Grafana alert flapping on bursty data windows	Set 15s evaluation + 1min resolution; added pending-period buffer
Go service goroutine leak under sustained load	Traced leak to dispatcher; fixed with ctx.Done() channel pattern
Metrics bridge cold-start delay (no data yet)	Pre-warm cache on startup; fall back to zero-values for missing metrics
Docker port conflicts on local dev machine	Used explicit bridge network + fixed port mappings in compose file
No LibreOffice on Mac for PPTX→PDF in QA	Used homebrew poppler (pdftoppm) + soffice for conversion pipeline

DataFlow Observatory — By the Numbers

~200/s Event Rate	**997K** Total Messages	**20K** DQ Checks Run	**10K+** Alerts Fired
250 Batch Size (events)	**5** dbt Models	**4** Alert Rules	**11** Grafana Panels
7 Docker Services	**4** Redis Key Types	**8** DQ Rule Checks	**3** Kafka Topics

Python 3.12 · Go 1.22 · Apache Kafka · Snowflake · dbt Core · Redis 7 · Prometheus · Grafana · Docker